

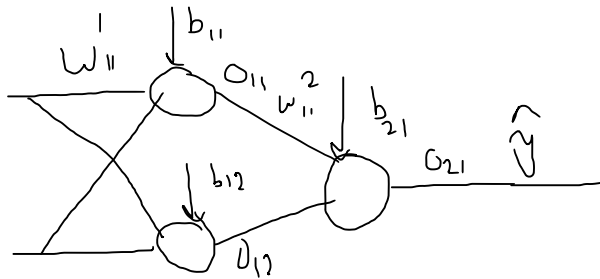
Recap

20 December 2025 21:37

Loss func = f (all trainable params) ✓

$$L = f(\hat{y}) = g(w, b)$$
$$= f(g(w, b))$$

$\left\{ \begin{array}{l} \Rightarrow \text{Sigmoid AF} \\ \rightarrow \text{Tanh AF} \end{array} \right. \quad \rightarrow \text{for deep neural network}$



$$w'_{11} = 1$$

$$\frac{dL}{dw'_{11}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{do_{21}} \cdot \frac{do_{21}}{dz_{21}} \cdot \frac{dz_{21}}{do_{11}} \cdot \frac{do_{11}}{dz_{11}} \cdot \frac{dz_{11}}{dw'_{11}}$$

individual derivatives value = < 1

0.1 0.1 0.1 0.1 0.1 0.1

$$\frac{dL}{dw'_{11}} = 0.000001$$

$$w_n = w_0 - \eta \frac{dL}{dw_0}$$

$$\eta = 1$$

$$w'_{11} = w_{11} - \eta \frac{dL}{dw'_{11}}$$

$$= 1 - 0.1 \times 0.000001$$

$$= 1 - 0.0000001$$

$$= 0.9999999 \approx 1$$

$$\sigma(w'_{11}) \approx \sigma(w_{11})$$

$$50\% \text{ VGP} = \text{Training}^X$$

Solutions?

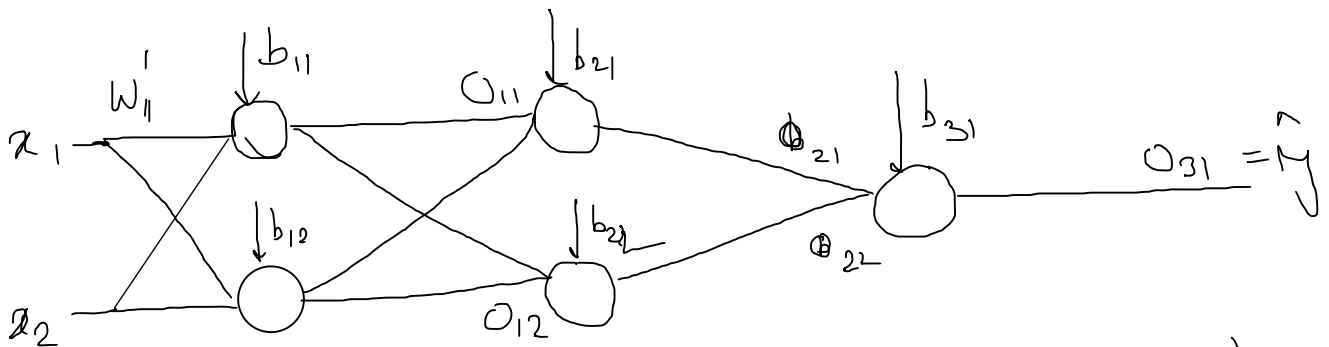
1. Shallow Neural Network.
2. change Act. func
3. Proper weight initialization
4. Batch Normalization.
5. Residual network.

$$w_n = w_0 - \eta \left[\frac{dL}{dw_0} \right]$$

$$\frac{dL}{dw_0} = \frac{w_0 - w_n}{\eta}$$

Sigmoid function
Tanh function

AF = Sigmoid



$$\frac{dL}{dw'_{11}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{dz_3} \cdot \frac{dz_3}{do_{21}} \cdot \boxed{\frac{do_{21}}{dz_{21}}}$$

$$o_{31} = \sigma(z_{31})$$

$$o_{21} = \sigma(z_{21})$$

$$o_{22} = \sigma(z_{22})$$

$$o_{11} = \sigma(z_{11})$$

$$o_{12} = \sigma(z_{12})$$

$$\frac{dz_{21}}{do_{11}} \cdot \boxed{\frac{do_{11}}{dw'_{11}}}$$

$$\sigma(z) = \frac{1}{1 + e^{-z}} = \begin{cases} 0 & (0.1) \\ 1 \end{cases}$$

$$\sigma'(z) = \sigma(z) \cdot (1 - \sigma(z)) = [0, 0.25]$$

0.1 (1 - 0.1)

$$= 0.1 \times 0.9$$

$$= \boxed{0.09}$$

Saturating function $\rightarrow$ VGP

$$\tanh(z)$$

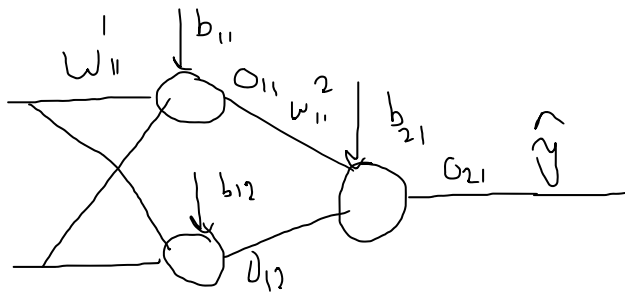
$$\tanh'(z) = \frac{1 - \tanh^2(z)}{1}$$

$$\frac{d \tanh \theta}{d \theta} = \underline{\underline{\sec^2 \theta}}$$

$$\begin{aligned} \sec^2 \theta + \tan^2 \theta &= 1 \\ \sec^2 \theta &= 1 - \tan^2 \theta \end{aligned}$$

Exploding Gradient Problem

22 December 2025 10:33



$$w'_{11} = 1$$

$$\frac{dL}{dw'_{11}} = \frac{dL}{d\hat{y}} \cdot \frac{d\hat{y}}{do_{21}} \cdot \frac{do_{21}}{dz_{21}} \cdot \frac{dz_{21}}{do_1} \cdot \frac{do_1}{dz_{11}} \cdot \frac{dz_{11}}{dw'_{11}}$$

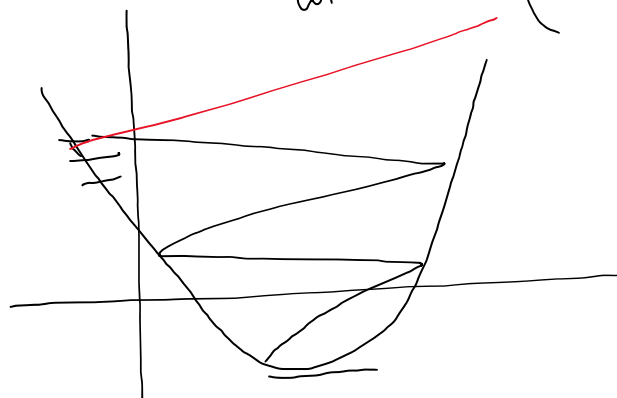
$$> 1$$

$$w_\eta = w_0 - \eta$$

$$\frac{dL}{dw_0}$$

Large Update

without η



$$\frac{dL}{dw_0}$$

sign